

Comparing Performance Patterns in Tested vs. Untested Powerlifting Competition Entries

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1. Introduction and Motivation

Few questions in strength sports generate more debate than whether an athlete competes with or without performance-enhancing drugs (PEDs), a topic commonly framed as "natty or not?" in gym culture. Competitive powerlifting provides a natural setting for studying this question with data, because federations and meets vary in whether they require drug testing. The OpenPowerlifting public dataset (OpenPowerlifting, 2026), containing over 3.8 million competition records, captures this distinction through a "Tested" field that marks whether a given entry counted as drug-tested.

Understanding how performance differs across these contexts could help athletes and coaches set realistic benchmarks for specific competition situations.

2. Dataset Description

2.1 Source

The OpenPowerlifting bulk CSV file is a public, community-maintained archive of powerlifting competition results. The raw file contains 3,868,570 rows and 42 columns, where each row is one competition entry (one lifter's result from one meet).

2.2 Key Features

Each row in the dataset is one competition entry. The analysis uses **Name** (to identify lifters), **Sex** (M/F), **Age** (on meet day; approximate ages appear as half-values like 23.5), **Equipment** (Raw, Wraps, Single-ply, or Multi-ply), and **WeightClassKg** to define subgroups. Performance is measured by **TotalKg** (sum of best squat, bench press, and deadlift) and **Dots** (a bodyweight-normalized score replacing the older Wilks formula; Kopayev et al., 2020). **Tested** marks whether the entry counted as drug-tested, not whether the lifter actually took a test. Four additional fields (**Event**, **Division**, **Place**, **AgeClass**) were used only during cleaning.

2.3 Preprocessing

Only full-power Open-division entries from male or female competitors in a standard equipment type (Raw, Wraps, Single-ply, Multi-ply) with a valid total were kept. Disqualifications, doping disqualifications, no-shows, and anyone under 18 were removed.

The **Tested** field was converted to a true/false flag, and lifters whose names appeared on both sides of the tested/non-tested divide (crossover lifters) were dropped so the two groups would not share members. After all filters, the cleaned dataset contained 510,882 rows, split roughly two-thirds tested and one-third non-tested.

Six age bins (18-23, 24-30, 31-40, 41-50, 51-60, 61+) were created for RQ3. Repeated entries by the same lifter within the same grouping were collapsed to a single representative row by averaging, producing three focused tables: one keyed by lifter, sex, equipment, and weight class (RQ1); one by lifter and sex (RQ2); and one by lifter, sex, and age bin (RQ3).

2.4 Challenges

Age is missing for 37.1% of raw rows (28.7% after cleaning), which significantly shrinks the sample available for RQ3. The oldest age bins (especially 61+) have the fewest rows, making their bootstrap ranges wider and their evidence less precise.

3. Research Questions / Objectives

RQ1: Main Performance Comparison

Do non-tested entries tend to show higher TotalKg and DOTS scores than tested entries? Comparisons are grouped by sex, weight class, and equipment so that each matchup is apples-to-apples. *H0: Non-tested entries do not have higher mean DOTS/TotalKg than tested entries within the same subgroup. Ha: Non-tested entries have higher mean DOTS/TotalKg (one-sided).*

RQ2: Tested-Entry Share by Performance Level

Is tested status associated with DOTS-score bin, separately by sex? Entries are divided into equal-width DOTS bins, and the tested-entry share is compared across bins. *H0: DOTS-score bin and tested status are independent. Ha: DOTS-score bin and tested status are associated within each sex.*

RQ3: Age Pattern Comparison

Do non-tested entries tend to show higher mean TotalKg across age bins, and how does that gap look at different ages? Rather than a formal hypothesis test, RQ3 uses bootstrap intervals to check whether the non-tested-minus-tested gap in each age bin is clearly above zero, treating the direction and size of the gap as the main evidence.

4. Methods / Approaches

All resampling procedures use a fixed random seed (`np.random.seed(604)`) for reproducibility. Multi-ply subgroups were excluded from the analysis because none met the minimum-count threshold of 30 rows on both sides of the tested/non-tested split.

4.1 RQ1 Methods

Using representative rows keyed by `Sex x Equipment x WeightClassKg x is_tested`, subgroups with fewer than 30 rows on either side are excluded. For each qualifying subgroup the non-tested-minus-tested gap is assessed three ways: **a 95% bootstrap CI** (10,000 resamples, percentile method); **a one-sided permutation test** (10,000 shuffles); and **a one-sided Welch t-test** as a parametric cross-check.

4.2 RQ2 Methods

Using representative rows keyed by **Sex x is_tested**, the representative DOTS values are split into five equal-width bins and a contingency table crosses bin by tested status. **A chi-squared test of independence** evaluates whether the tested-entry share differs across bins.

4.3 RQ3 Methods

Using representative rows keyed by **Sex x age_bin x is_tested**, a **KDE overlay plot** (Silverman bandwidth) compares the full TotalKg distributions for tested and non-tested groups in two representative bins (24-30 and 41-50). For each **Sex x age_bin** pair a **bootstrap resampling** (10,000 resamples) computes the non-tested-minus-tested gap in mean TotalKg and records the 5th and 95th percentiles; when the 5th percentile is above zero the bin supports higher mean TotalKg for non-tested rows.

5. Results and Analysis

5.1 RQ1 - Main Performance Comparison

Among the 78 subgroup pairs that met the minimum-count rule, the overall pattern was mixed because equipment types pulled in opposite directions (Table 1).

Table 1. RQ1 bootstrap 95% CI results by equipment type (78 qualifying subgroup pairs)

Equipment	Non-tested higher	Tested higher	No clear difference	Subgroups
Raw	14 (37%)	13 (34%)	11 (29%)	38
Wraps	19 (95%)	0 (0%)	1 (5%)	20
Single-ply	1 (5%)	19 (95%)	0 (0%)	20
All	34 (44%)	32 (41%)	12 (15%)	78

RQ1: Mean DOTS by Equipment Type and Tested Status

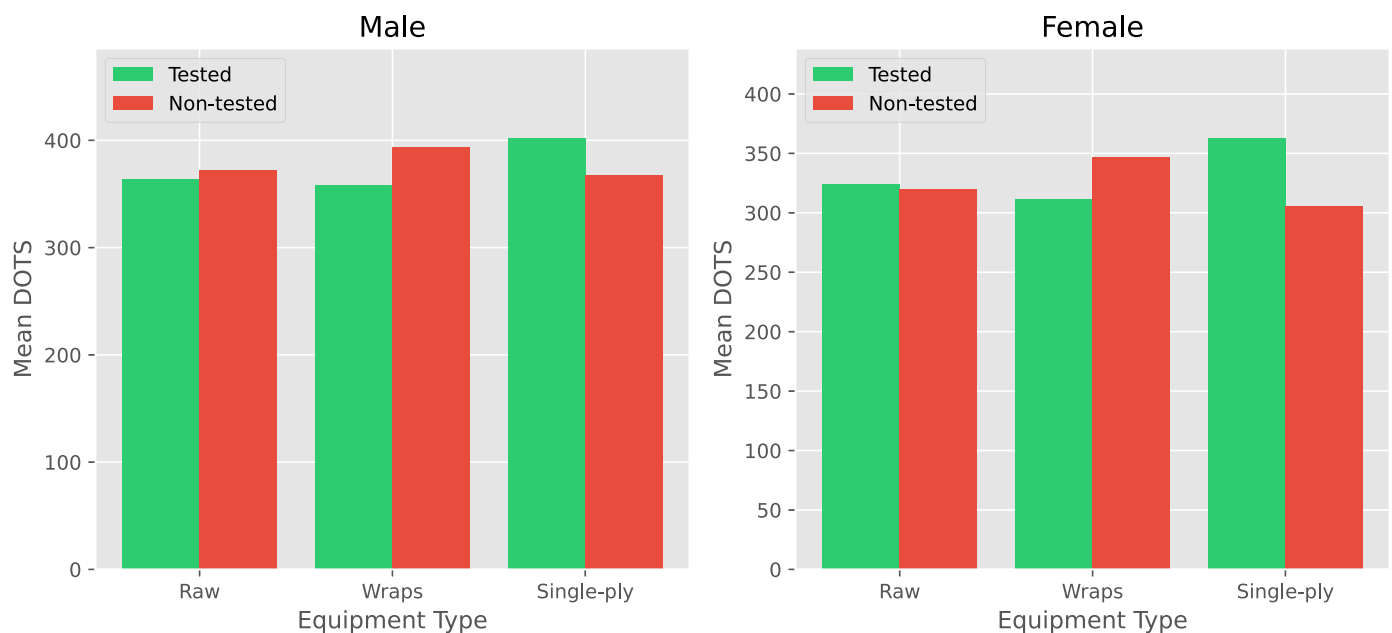


Figure 1. Overall mean DOTS by equipment type and tested status, shown separately for men (left) and women (right). Green bars represent tested entries; red bars represent non-tested entries.

Figure 1 shows the same split in plain form: the non-tested bar is visibly taller in Wraps, shorter in Single-ply, and nearly even in Raw. Multi-ply is absent because none of its subgroups met the minimum-count rule. The permutation test and Welch t-test agreed with the bootstrap CI directions across all subgroups. Among subgroups with a clear non-tested advantage, the TotalKg gap ranged from about 6 kg to 85 kg, with a median around 35 kg.

5.2 RQ2 - Tested Status by DOTS-Score Bin

Table 2. RQ2 chi-squared results by sex

Sex	Chi-squared statistic	df	p-value	Expected-count check	Interpretation
Men	166.99	4	< 0.001	All ≥ 5 (min 83.46)	Reject H0
Women	510.37	4	< 0.001	All ≥ 5 (min 7.93)	Reject H0

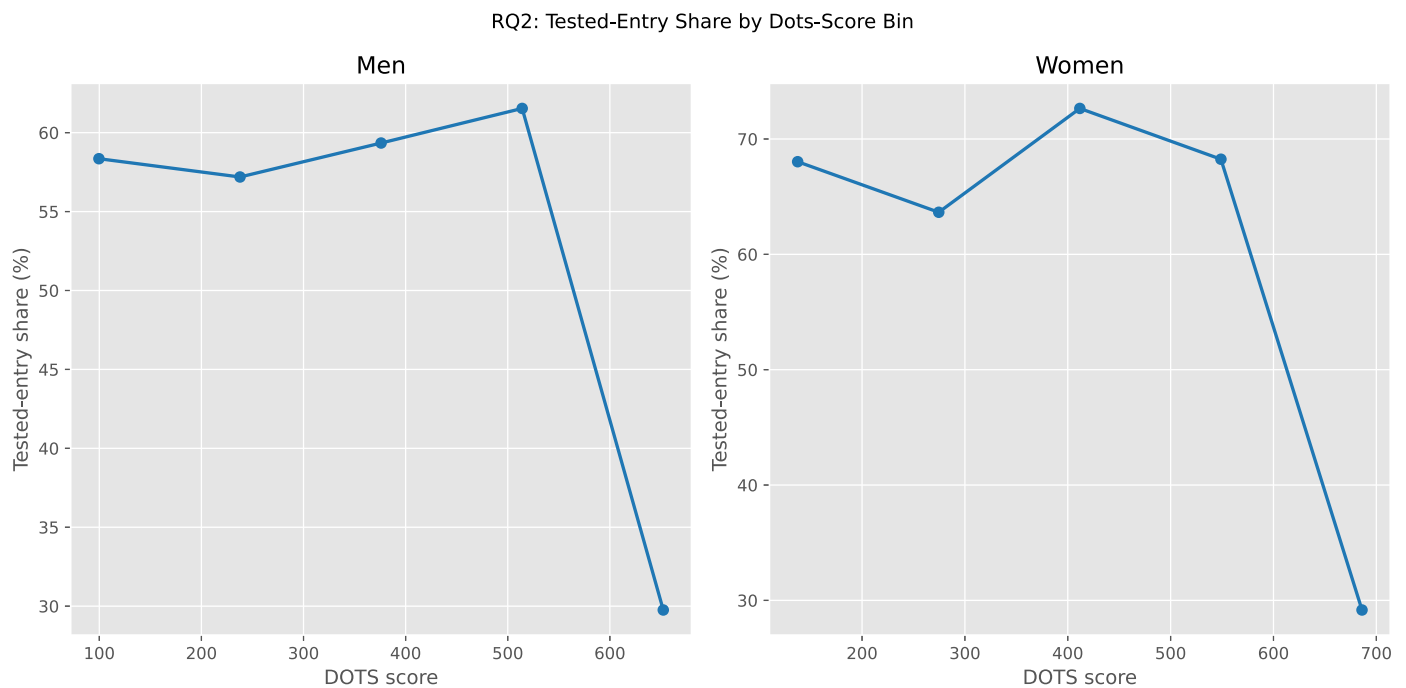


Figure 2. Tested-entry share (%) by Dots-score bin, shown separately for men (left) and women (right).

The pattern was concentrated at the top of the scoring range rather than spread evenly across all bins. For men, the tested-entry share hovers in a fairly narrow band (~57-62%) across the first four bins, then drops sharply to about 30% in the highest-DOTS bin. For women, the share varies more across the first four bins (~64-73%) but shows a similar drop to about 29% at the top. In plain terms, the top bin was the only place where non-tested entries outnumbered tested entries.

5.3 RQ3 - Age Pattern Comparison

The pattern depended on both sex and age, as Figure 3 illustrates.

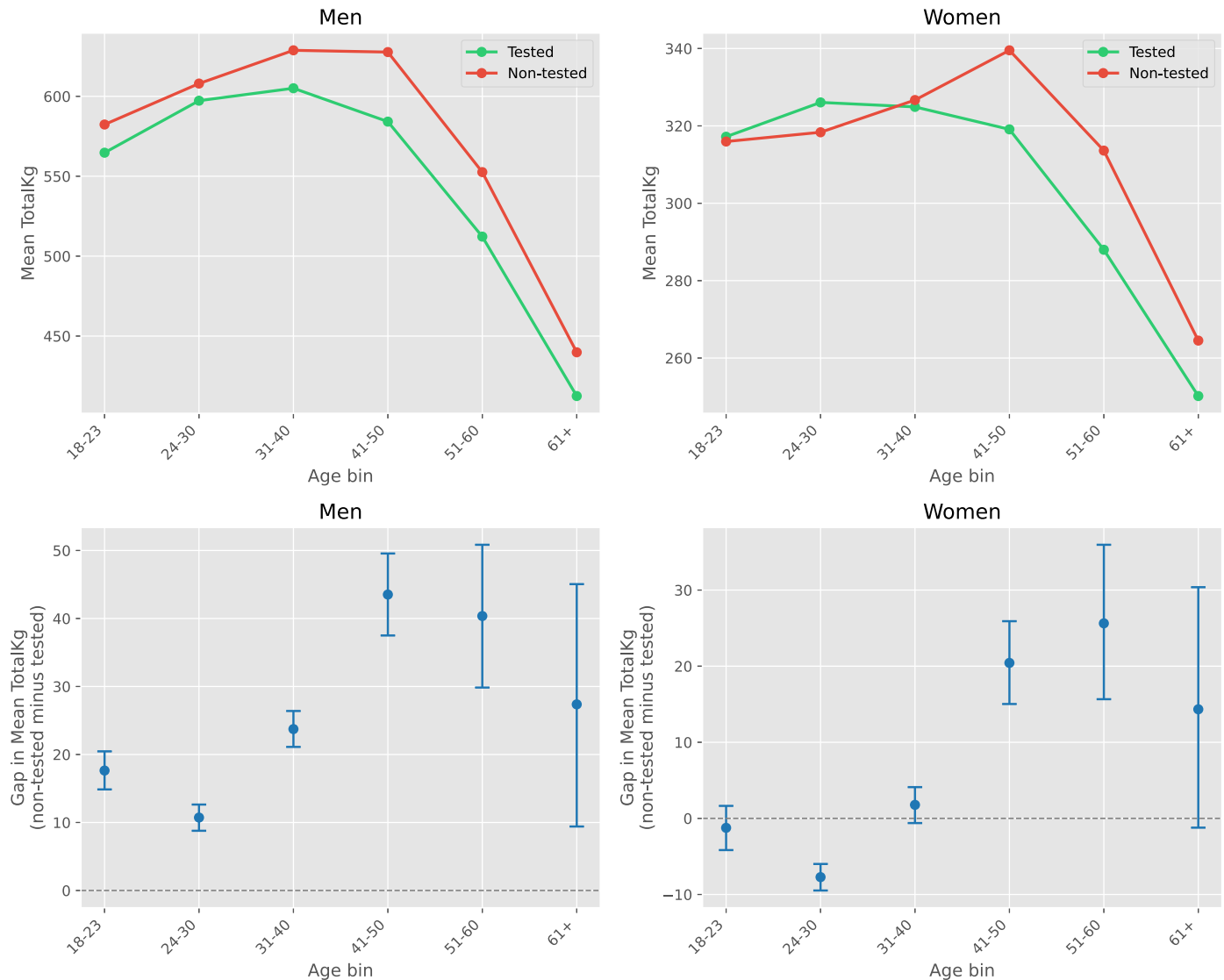


Figure 3. Mean TotalKg by age bin and tested status. Men are shown on the left, women on the right.

Figure 3 shows a split story. For men, every age bin favored non-tested entries, but the gap varied in size: smallest in the 20s (~11 kg), widening through middle age, and peaking around 41-50 (~44 kg). For women, the younger bins stayed mixed or tilted toward tested entries, while the 41-50 and 51-60 bins showed a clearer non-tested advantage (~20-26 kg). The oldest bins (especially 61+) had the widest whiskers, reflecting smaller sample sizes.

6. Discussion / Insights

6.1 Significance

No single narrative fits the full picture. The opposite equipment-type patterns hint that the competitive landscape of each equipment category may matter as much as the testing policy itself. The concentration of top DOTS scores in non-tested contexts may reflect that lifters chasing the highest totals tend to gravitate toward non-tested federations, which are often larger and more competitive.

6.2 Limitations

As noted in Section 2.2, the "Tested" field is a competition-context label, not proof of drug use or non-use. Athletes choose which context to compete in for many reasons, so the data cannot separate one explanation from another.

The cleaning described in Section 2.3 retained about 13% of rows and removed all crossover lifters, so the findings describe a specific slice of the sport, not all powerlifters; if the excluded entries follow different patterns, the conclusions may not generalize. Because crossover detection used name alone, some non-crossover lifters who share a name may have been incorrectly removed.

Finally, older meets tend to have more missing fields, and federation conventions around weight classes and equipment categories were not normalized. Some of the observed patterns may partly reflect which federations contributed the most data rather than a true tested/non-tested difference.

6.3 Improvements / Extensions

Three extensions could build on these findings. First, a time-based analysis could track whether the tested/non-tested gap has widened, narrowed, or stayed stable over the years. Second, breaking the analysis down by federation could reveal whether the patterns are driven by a few large organizations or hold broadly across the sport. Third, a follow-up study could focus on the crossover athletes excluded from this project and compare their performance across contexts, using each lifter as their own control to reduce the self-selection problem discussed in Section 6.2.

7. Use of AI Tools

I used Claude (Anthropic) mainly as a learning and editing assistant. It helped me understand unfamiliar Python tools, troubleshoot notebook errors, proofread drafts, and sketch a starter outline for the report. For example, it helped me understand tools like `pd.cut()` for age bins and the `alternative` argument in `scipy.stats.ttest_ind()` for one-sided tests. I chose the project topic, designed the research questions, and made the analysis decisions myself. When Claude suggested ideas outside the scope of the course, I either rejected them or adapted them to match class methods. The notebook code and report were AI-assisted, but ultimately written by me.

8. Conclusion

This project set out to ask whether non-tested powerlifting competition entries tend to show higher performance than tested entries. The short answer is: **it depends**.

RQ1 found that the non-tested advantage depended heavily on equipment type, appearing consistently in Wraps but reversing in Single-ply. RQ2 found that non-tested entries only outnumbered tested entries in the highest-scoring DOTS bin, with little separation elsewhere. RQ3 found that the gap was widest in middle and older age bins and much clearer for men than for women.

The caveats in Section 6.2 apply throughout. The dramatically opposite patterns across equipment types suggest that competitive context may shape these results as much as testing policy does, and the results should not be read as a simple story about drug use.

References

Kopayev, O., Onyshchenko, B., & Stetsenko, A. (2020). *Report: Evaluation of Wilks, Wilks-2, DOTS, IPF and Goodlift formulas for calculating relative scores in IPF powerlifting competitions*. International Powerlifting Federation.
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